

DATA ANALYSIS EXAMINATION

Maximum score : 80

Duration : 2 Hours

PART A (All questions carry marks 2.5 marks)

1. What is EDA? Explain.
2. What is the purpose of scaling a dataset? Why is it important to scale data before training a machine learning model?
3. Explain how the MinMaxScaler works. What range does it scale the data to?
4. What is the main advantage of using RobustScaler over StandardScaler?
5. What is one-hot encoding? Provide an example of how you would apply one-hot encoding to a "Color" column containing the values ["Red", "Blue", "Green"].
6. What is label encoding? How does it work, and when should it be used instead of onehot encoding?
7. What are outliers, and why are they a problem in data analysis and machine learning models?
8. If an outlier is detected in a dataset, what are the different ways to handle it? Mention at least two methods.

PART B (All questions carry marks 12 marks)

❖ Load the Dataset: (file:"Loan_data")

- Load the dataset
- Display column names and dataset shape.
- For the column "ApplicantIncome", compute and explain:
 - ◆ Mean, Median, Mode
 - ◆ Standard Deviation, Variance
 - ◆ Skewness and Kurtosis

❖ Handling Missing Data:

- Check for missing values in each column. If any column has missing values, how would you handle them (e.g., drop or impute)?
- Implement mean, median, and mode imputation for missing values in the dataset. How do they affect your data distribution?

❖ Outlier Detection:

- Use boxplots to detect outliers in numerical columns.
- Identify and remove outliers.
- Visualize the impact of removing outliers on the dataset.

❖ Encoding Categorical Features:

- Apply Label Encoding to a categorical column such as “Property_Area”.
- Perform One-Hot Encoding on the categorical column “Married” & “Gender”.
What changes in the dataset after encoding?
- If a column has ordered categories, like “Education”, apply Ordinal Encoding.
How do the results compare to other encoding methods?

❖ Scaling and Normalization:

- Apply Min-Max Scaling to all columns. How does scaling impact your data?
- Standardize the columns using Z-score normalization and compare the results.
- Apply Robust Scaling to deal with outliers in columns that may have extreme values.